

Program Summary - THE4.sas

Execution Environment

Author: u64127570
File: /home/u64127570/sasuser.v94/THE4.sas
SAS Platform: Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host: ODAWS01-USW2-2.ODA.SAS.COM
SAS Version: 9.04.01M7P08062020
SAS Locale: en_US
Submission Time: 4/17/2025, 10:40:00 PM
Browser Host: C-67-165-187-91.HSD1.IL.COMCAST.NET
User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0
Application Server: ODAMID00-USW2-2.ODA.SAS.COM

Code: THE4.sas

```
/* Take Home Exam Q4 */
```

Log: THE4.sas

Notes (27)

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69      /* Take Home Exam Q4 */
70
71      proc format;
72          value $sexf
73              'M' = 'male'
74              'F' = 'female';
NOTE: Format $SEXF is already on the library WORK.FORMATS.
NOTE: Format $SEXF has been output.
75          value $usef
76              0 = 'does not use seatbelt'
77              1 = 'uses a seatbelt';
NOTE: Format $USEF is already on the library WORK.FORMATS.
NOTE: Format $USEF has been output.
78          value $locationf
79              0 = 'rural'
80              1 = 'urban';
NOTE: Format $LOCATIONF is already on the library WORK.FORMATS.
NOTE: Format $LOCATIONF has been output.
81      run;
```

NOTE: PROCEDURE FORMAT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	246.59k

```

OS Memory          24736.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         67  Switch Count  0
Page Faults        0
Page Reclaims      14
Page Swaps         0
Voluntary Context Switches  0
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations  32

```

```

82
83      Proc import datafile='/home/u64127570/sasuser.v94/SeatBelts2025.xlsx'
84      Out=belts DBMS=  xlsx Replace;
85      Getnames=no;
86      run;

```

NOTE: One or more variables were converted because the data type is not supported by the V9 engine. For more details, run with options MSGLEVEL=I.

NOTE: The import data set has 300 observations and 1 variables.

NOTE: WORK.BELTS data set was successfully created.

NOTE: PROCEDURE IMPORT used (Total process time):

```

real time          0.01 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            3362.28k
OS Memory          27384.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         68  Switch Count  4
Page Faults        0
Page Reclaims      695
Page Swaps         0
Voluntary Context Switches  31
Involuntary Context Switches 2
Block Input Operations  0
Block Output Operations  272

```

```

87
88      data use_belt;
89          set belts;
90          sex = substr(A, 1, 1);
91          use = substr(A, 2, 1);
92          location = substr(A, 3, 1);
93          format sex sexf. use usef. location locationf.;
94      run;

```

NOTE: There were 300 observations read from the data set WORK.BELTS.

NOTE: The data set WORK.USE_BELT has 300 observations and 4 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            952.65k
OS Memory          25256.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         69  Switch Count  2
Page Faults        0
Page Reclaims      118
Page Swaps         0
Voluntary Context Switches  10
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations  264

```

```

95
96      proc sgplot data = use_belt;
97          title "Seatbelt Usage by Sex";
98          vbar use /group=sex ;
99          xaxis label= "Seatbelt Use"; *"use" doesnt make a great label;
100      run;

```

NOTE: PROCEDURE SGPLOT used (Total process time):

```

real time          0.09 seconds
user cpu time      0.03 seconds
system cpu time    0.01 seconds
memory            8513.71k
OS Memory          30252.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         70  Switch Count  3
Page Faults        0
Page Reclaims      1753
Page Swaps         0
Voluntary Context Switches  187

```

```
Involuntary Context Switches      0
Block Input Operations             0
Block Output Operations           608
```

NOTE: There were 300 observations read from the data set WORK.USE_BELT.

```
101
102      /* Seatbelt use is less frequent than not using a seatbelt, overall and by sex. Out of those who use a seatbelt, a
102      ! greater proportion are female. of those who do not use a seatbelt, distrubtion of sexes looks about equal.
103      there are more females than males in this data */
104
105      proc freq data= use_belt;
106          tables sex*use / chisq relrisk;
107      run;
```

NOTE: There were 300 observations read from the data set WORK.USE_BELT.

```
NOTE: PROCEDURE FREQ used (Total process time):
real time          0.04 seconds
user cpu time      0.05 seconds
system cpu time    0.00 seconds
memory            2179.03k
OS Memory          30380.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         71  Switch Count  4
Page Faults        0
Page Reclaims      300
Page Swaps         0
Voluntary Context Switches  28
Involuntary Context Switches  4
Block Input Operations  0
Block Output Operations  528
```

```
108
109      /* null: there is no difference in seatbelt use and sex
110      alt: there is a difference in seatbelt use and sex
111      chisq: 3.1208 p= 0.0773 > 0.05
112      at a 95% confidence level, we fail to reject the null hypothesis, there is significant evidence that there is no
112      ! difference in seatbelt use and sex
113      odds ratio: 0.4108, 1.0482
114      the odds ratio confidence limits include 1, indicating that there is no signifcant association between sex and seatbelt
114      ! use. */
115
116      proc sgplot data = use_belt;
117          title "Seatbelt Usage by Location";
118          vbar use /group=location ;
119          xaxis label= "Seatbelt Use"; /*"use" doesnt make a great label;
120      run;
```

```
NOTE: PROCEDURE SGPLOT used (Total process time):
real time          0.08 seconds
user cpu time      0.03 seconds
system cpu time    0.01 seconds
memory            2330.62k
OS Memory          31408.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         72  Switch Count  3
Page Faults        0
Page Reclaims      613
Page Swaps         0
Voluntary Context Switches  183
Involuntary Context Switches  3
Block Input Operations  0
Block Output Operations  376
```

NOTE: There were 300 observations read from the data set WORK.USE_BELT.

```
121
122      /* Seatbelt use is less frequent than not using a seatbelt overall and for each location. Out of those who use a
122      ! seatbelt, a greater proportion are urban. of those who do not use a seatbelt, a greater proportion is urban.
123      There is more urban data than rural*/
124
125      proc freq data= use_belt;
126          tables location*use / chisq;
127      run;
```

NOTE: There were 300 observations read from the data set WORK.USE_BELT.

```
NOTE: PROCEDURE FREQ used (Total process time):
real time          0.03 seconds
user cpu time      0.04 seconds
system cpu time    0.00 seconds
memory            1322.87k
OS Memory          30380.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         73  Switch Count  4
Page Faults        0
```

```

Page Reclaims          185
Page Swaps              0
Voluntary Context Switches 26
Involuntary Context Switches 1
Block Input Operations   0
Block Output Operations  528

128
129 /* null: there is no difference in seatbelt use and location
130 alt: there is a difference in seatbelt use and location
131 chisq: 1.3690 p=0.2420 >0.05
132 at a 95% confidence level, we fail to reject the null hypothesis, there is significant evidence that there is no
133 ! difference in seatbelt use and location */
134 proc sort data= use_belt;
135 by location;
136 run;

NOTE: There were 300 observations read from the data set WORK.USE_BELT.
NOTE: The data set WORK.USE_BELT has 300 observations and 4 variables.
NOTE: PROCEDURE SORT used (Total process time):
real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory             963.25k
OS Memory          30120.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         74 Switch Count  2
Page Faults        0
Page Reclaims      113
Page Swaps         0
Voluntary Context Switches 13
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 264

137
138 proc freq data = use_belt;
139 title "Seatbelt Use and Sex by Location";
140 tables use*sex / chisq relrisk expected;
141 by location;
142 run;

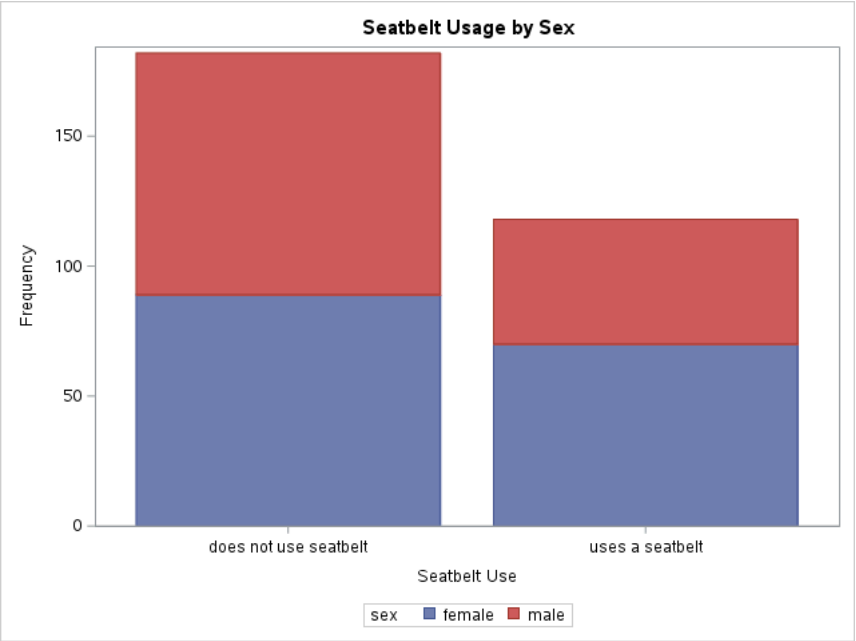
NOTE: There were 300 observations read from the data set WORK.USE_BELT.
NOTE: PROCEDURE FREQ used (Total process time):
real time          0.06 seconds
user cpu time      0.07 seconds
system cpu time    0.00 seconds
memory             1531.96k
OS Memory          30636.00k
Timestamp          04/18/2025 03:40:00 AM
Step Count         75 Switch Count  5
Page Faults        0
Page Reclaims      217
Page Swaps         0
Voluntary Context Switches 38
Involuntary Context Switches 2
Block Input Operations 0
Block Output Operations 560

143
144 /* null: there is no difference in seatbelt use and sex
145 alt: there is a difference in seatbelt use and sex
146 rural: chisq= 0.0450 p=0.8319
147 odds ratio= 0.4123, 2.0400
148 We fail to reject the null hypothesis, there is significant evidence that there is no difference in seatbelt use and sex.
149
150 urban: chisq= 5.2521 p = 0.0219
151 odds ratio= 0.2712, 0.9074
152 We reject the null hypothesis, there is significant evidence that there is a difference in seatbelt use and sex.
153
154 There is a difference in seatbelt use and sex by location.
155 In rural areas, 45.24% of females use a seatbelt and 43.10% of males use a seatbelt.
156 In urban areas, 43.59% of females use a seatbelt and 27.71% of males use a seatbelt.
157 The significant disparity by location is seen in males, with a 15.39% difference in males who use a seatbelt.
158 Location=rural has an odds ratio that contains 1, indication no significant relationship between seatbelt use and sex.
158 ! location=urban however, has an odds ratio that does not contain 1, indication a significant association between seatbeat
158 ! use and sex.
159 These odds ratios are visible in the above percents.
160
161 Simpsons paradox occurs when the total association between 2 categorical variables is qualitatively different than the
161 ! partial association between those variables after controlling another variable.
162 In this example, the total associations are demonstrated by seatbelt use and sex AND seatbelt use and location. Without

```

```
162 ! controlling the third variable, both of these tests showed no significant association.
163 However, once the third variable was controlled, location in this example, the true associations were able to be seen.
164 For rural areas, there is no association between seatbelt use and sex, resulting in a very high p value.
165 In urban area, there is a significant association between seatbelt use and sex, but when looked at as a whole, it's
165 ! significant p value is overcome by the rural areas, resulting in a no significance conclusion.
166 Without considering Simpons paradox and uncontrolled/unobserved variables, misleading conclusions on association can be
166 ! presented as fact, leading to misinformation and even discrimination, as demonstrated by the widely circulated "13% of
166 ! the population, 50% of crime" statistic that many use to justify their prejudice.
167 */
168
169
170
171
172 OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
182
```

Results: THE4.sas



Seatbelt Usage by Sex

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of sex by use			
	sex	use		Total
		does not use seatbelt	uses a seatbelt	
	female	89 29.67 55.97 48.90	70 23.33 44.03 59.32	159 53.00
	male	70 31.00 65.96 51.10	48 16.00 34.04 40.68	118 47.00
	Total	159 60.67	118 39.33	300 100.00

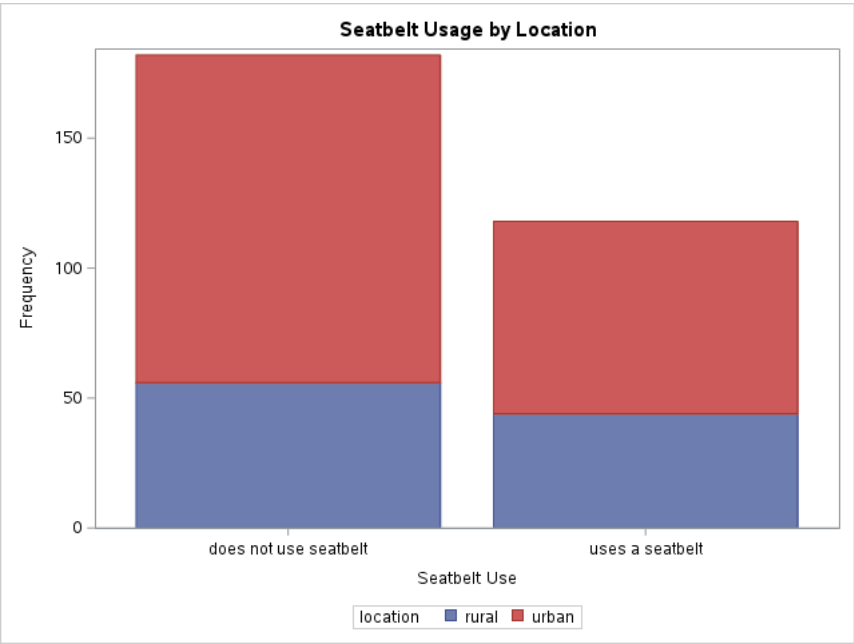
Statistics for Table of sex by use

Statistic	DF	Value	Prob
Chi-Square	1	3.1208	0.0773
Likelihood Ratio Chi-Square	1	3.1331	0.0767
Continuity Adj. Chi-Square	1	2.7165	0.0993
Mantel-Haenszel Chi-Square	1	3.1104	0.0778
Phi Coefficient		-0.1020	
Contingency Coefficient		0.1015	
Cramer's V		-0.1020	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	89
Left-sided Pr <= F	0.0495
Right-sided Pr >= F	0.9705
Table Probability (P)	0.0200
Two-sided Pr <= P	0.0972

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	0.6562	0.4108	1.0482
Relative Risk (Column 1)	0.8487	0.7076	1.0179
Relative Risk (Column 2)	1.2932	0.9687	1.7265

Sample Size = 300



Seatbelt Usage by Location

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of location by use			
	use			Total
	location	does not use seatbelt	uses a seatbelt	
rural		56	44	100
		18.67	14.67	33.33
		56.00	44.00	
		30.77	37.29	
urban		126	74	200
		42.00	24.67	66.67
		63.00	37.00	
		69.23	62.71	
Total		182	118	300
		60.67	39.33	100.00

Statistics for Table of location by use

Statistic	DF	Value	Prob
Chi-Square	1	1.3690	0.2420
Likelihood Ratio Chi-Square	1	1.3612	0.2433
Continuity Adj. Chi-Square	1	1.0913	0.2962
Mantel-Haenszel Chi-Square	1	1.3644	0.2428
Phi Coefficient		-0.0676	
Contingency Coefficient		0.0674	
Cramer's V		-0.0676	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	56
Left-sided Pr <= F	0.1481
Right-sided Pr >= F	0.9021
Table Probability (P)	0.0503
Two-sided Pr <= P	0.2606

Sample Size = 300

Seatbelt Use and Sex by Location

The FREQ Procedure

location=rural

Frequency Expected Percent Row Pct Col Pct	Table of use by sex			
	use	sex		
		female	male	Total
	does not use seatbelt	23	33	56
		23.52	32.48	56.00
		23.00	33.00	
		41.07	58.93	
		54.76	56.90	
	uses a seatbelt	19	25	44
		18.48	25.52	44.00
		19.00	25.00	
		43.18	56.82	
		45.24	43.10	
	Total	42	58	100
		42.00	58.00	100.00

Statistics for Table of use by sex

Statistic	DF	Value	Prob
Chi-Square	1	0.0450	0.8319
Likelihood Ratio Chi-Square	1	0.0450	0.8320
Continuity Adj. Chi-Square	1	0.0001	0.9935
Mantel-Haenszel Chi-Square	1	0.0446	0.8327
Phi Coefficient		-0.0212	
Contingency Coefficient		0.0212	
Cramer's V		-0.0212	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	23
Left-sided Pr <= F	0.4962
Right-sided Pr >= F	0.6617
Table Probability (P)	0.1579
Two-sided Pr <= P	0.8413

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	0.9171	0.4123	2.0400
Relative Risk (Column 1)	0.9511	0.5993	1.5094
Relative Risk (Column 2)	1.0371	0.7398	1.4540

Sample Size = 100

Seatbelt Use and Sex by Location

The FREQ Procedure

location=urban

Frequency Expected Percent Row Pct Col Pct	Table of use by sex			
	use	sex		
		female	male	Total
	does not use seatbelt	66	60	126
		73.71	52.29	63.00
		33.00	30.00	
		52.38	47.62	
		56.41	72.29	
	uses a seatbelt	51	23	74
		43.29	30.71	37.00
		25.50	11.50	
		68.92	31.08	
		43.59	27.71	
	Total	117	83	200
		58.50	41.50	100.00

Statistics for Table of use by sex

Statistic	DF	Value	Prob
Chi-Square	1	5.2521	0.0219
Likelihood Ratio Chi-Square	1	5.3408	0.0208
Continuity Adj. Chi-Square	1	4.5930	0.0321
Mantel-Haenszel Chi-Square	1	5.2258	0.0223
Phi Coefficient		-0.1621	
Contingency Coefficient		0.1600	
Cramer's V		-0.1621	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	66
Left-sided Pr <= F	0.0155
Right-sided Pr >= F	0.9930
Table Probability (P)	0.0086
Two-sided Pr <= P	0.0259

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	0.4961	0.2712	0.9074
Relative Risk (Column 1)	0.7600	0.6062	0.9529
Relative Risk (Column 2)	1.5321	1.0419	2.2528

Sample Size = 200